

Week: 11

Experiment: 10

“To prepare glass slides of onion root tips to study stages of mitosis.”

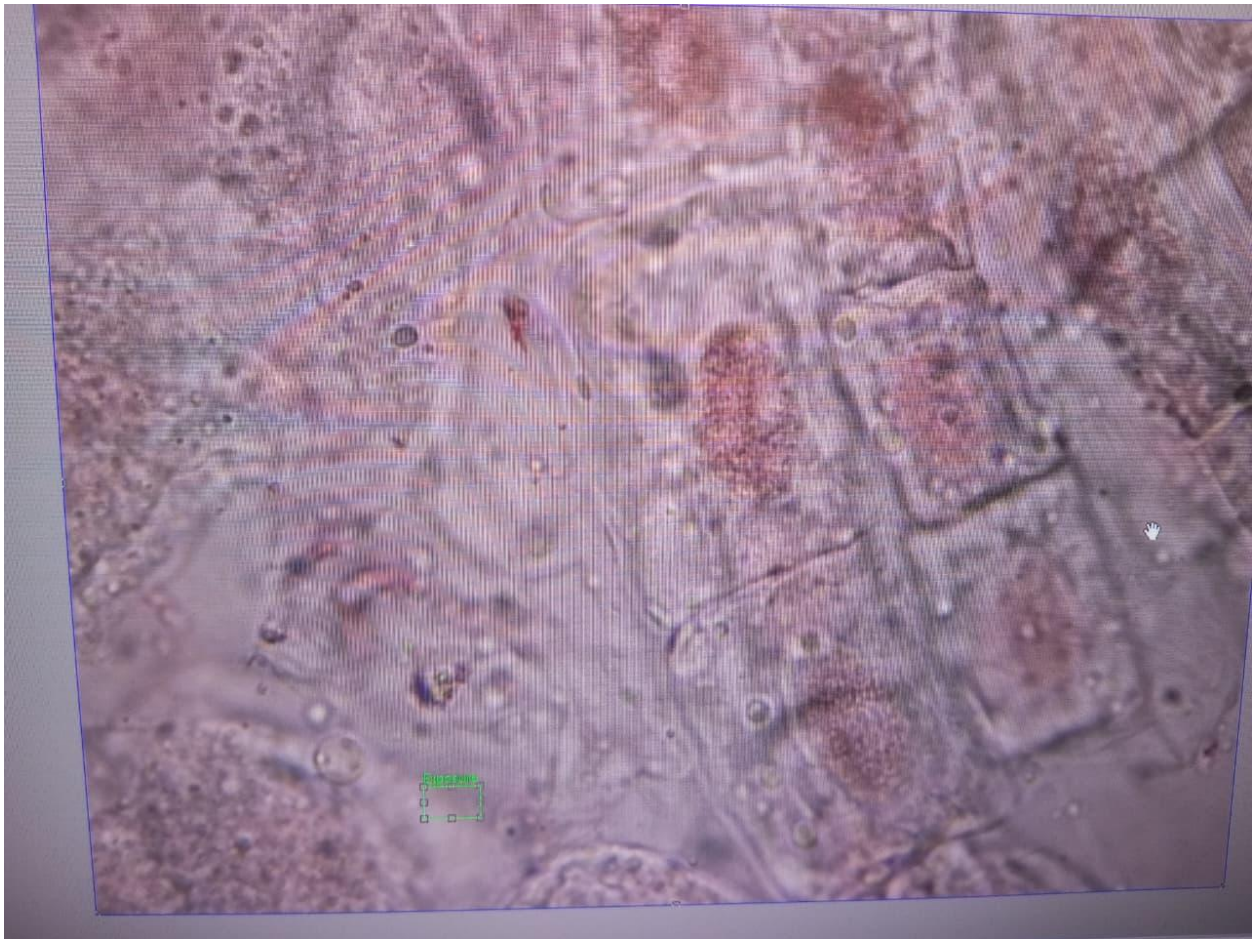
1) Lab Performance (10 Marks):

2) Total Marks (20 Marks):

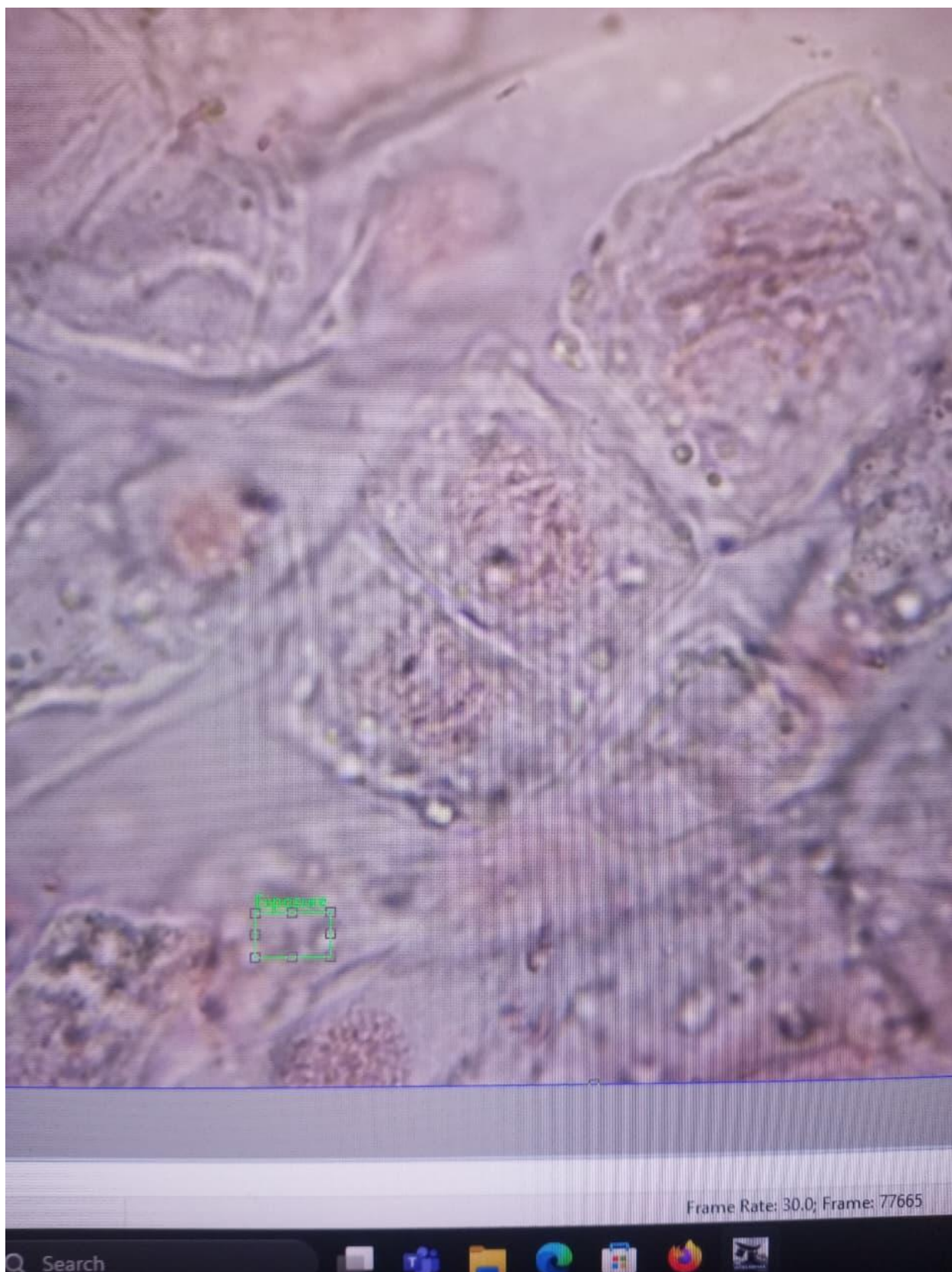
3) Observation (5 marks):

(Paste 5 Pictures of different stages of mitosis observed in the microscope.)

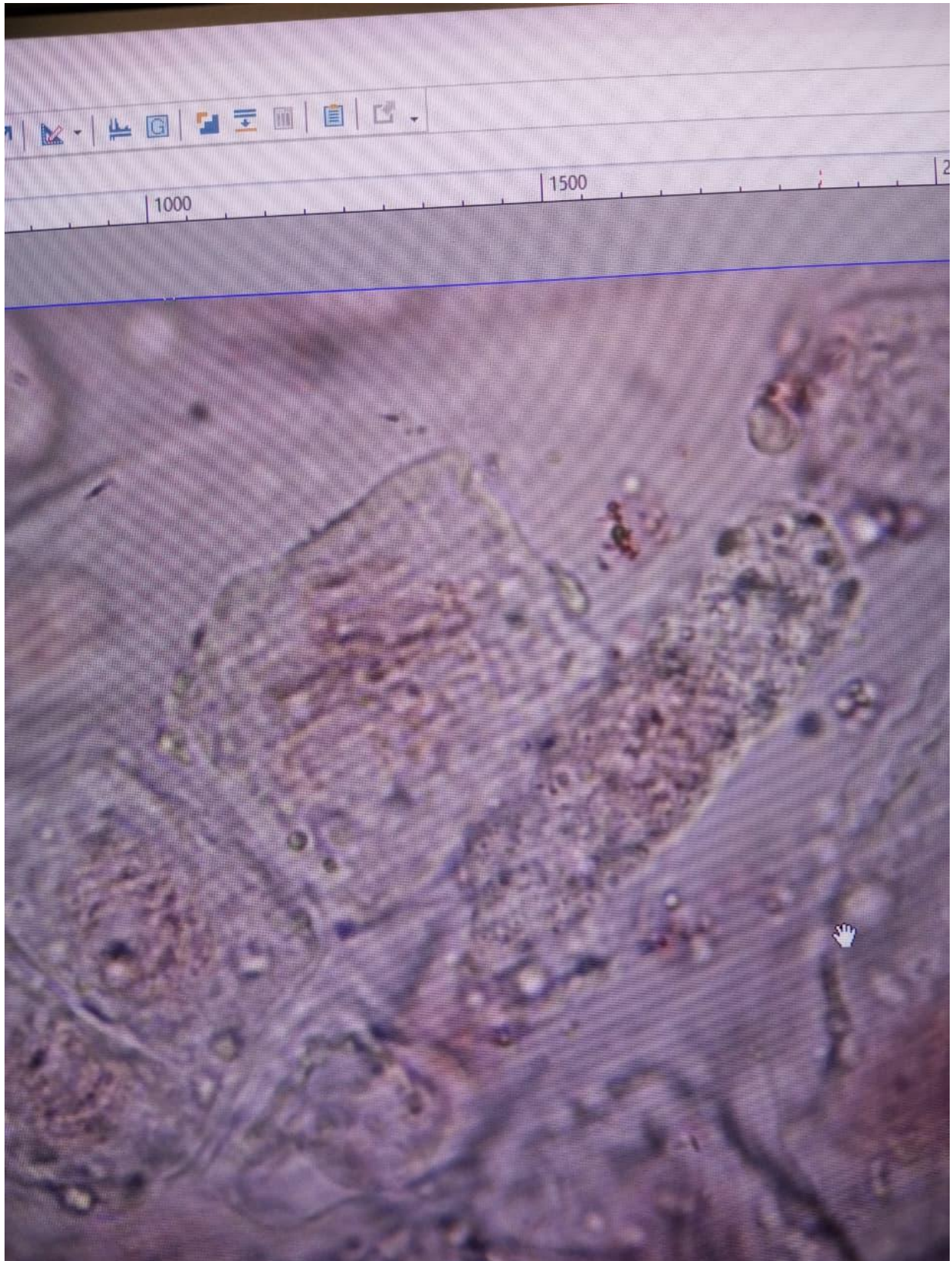
INTERPHASE:-



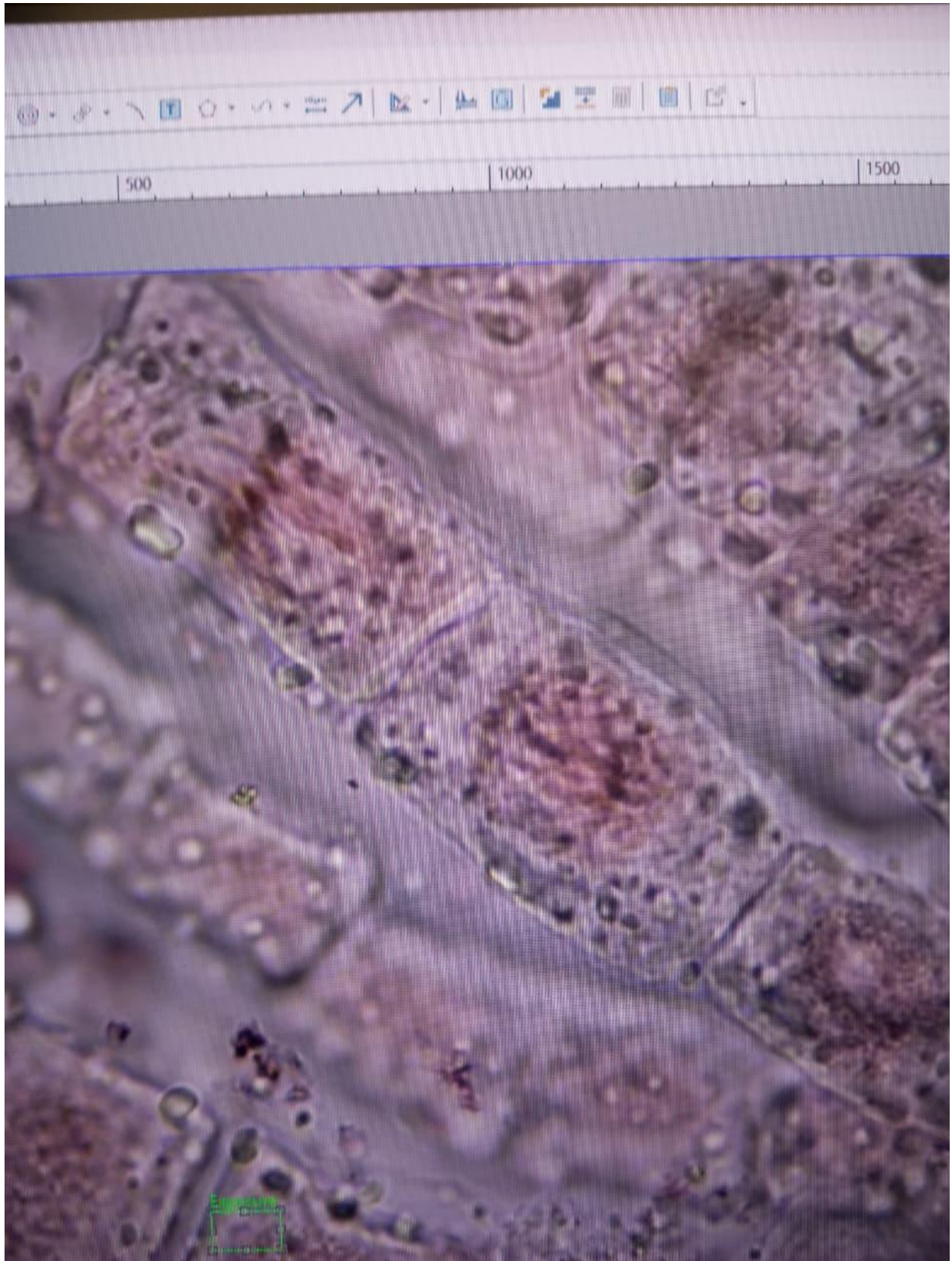
CYTOKYNEOSIS:-



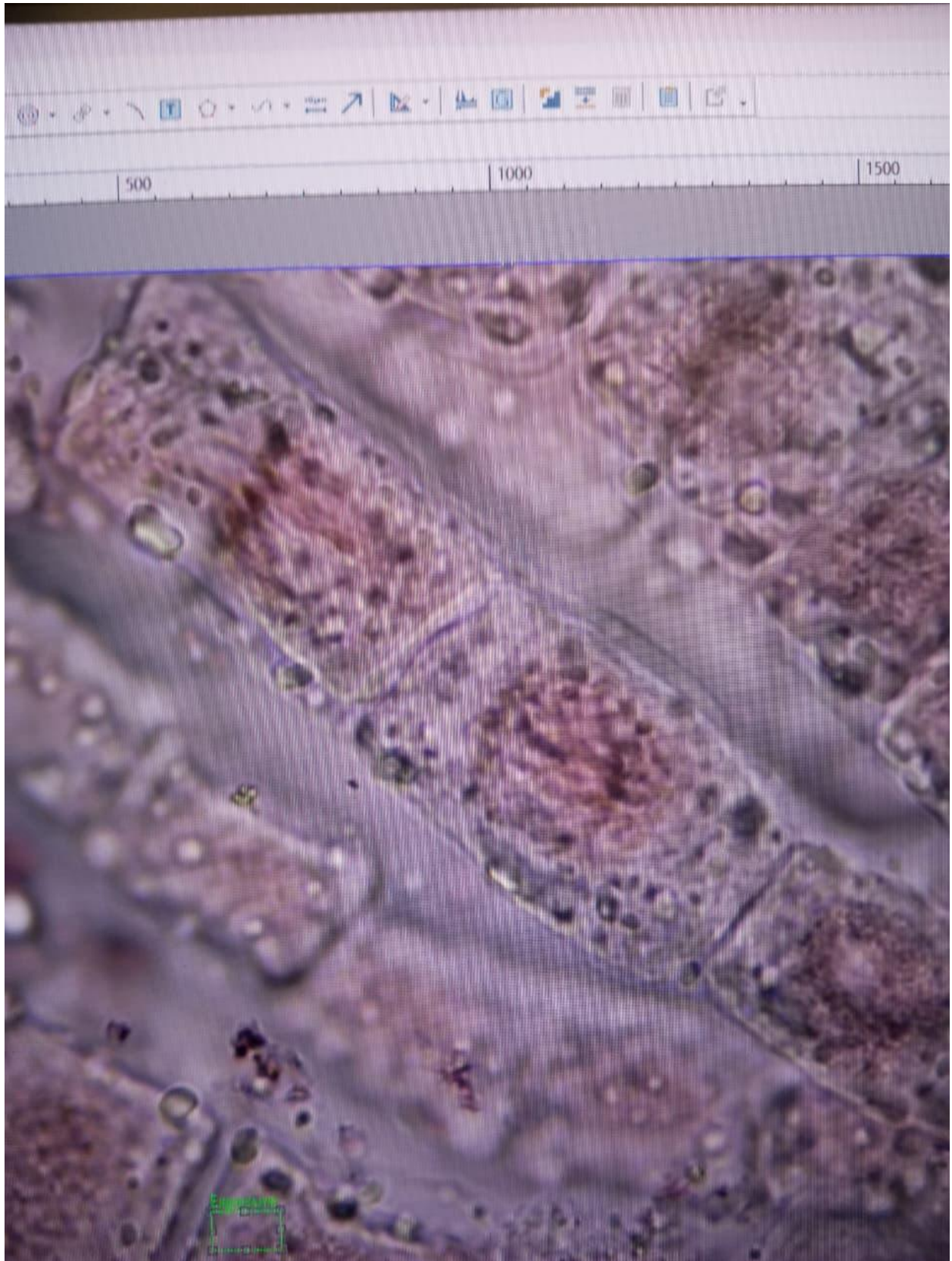
METAPHASE:-



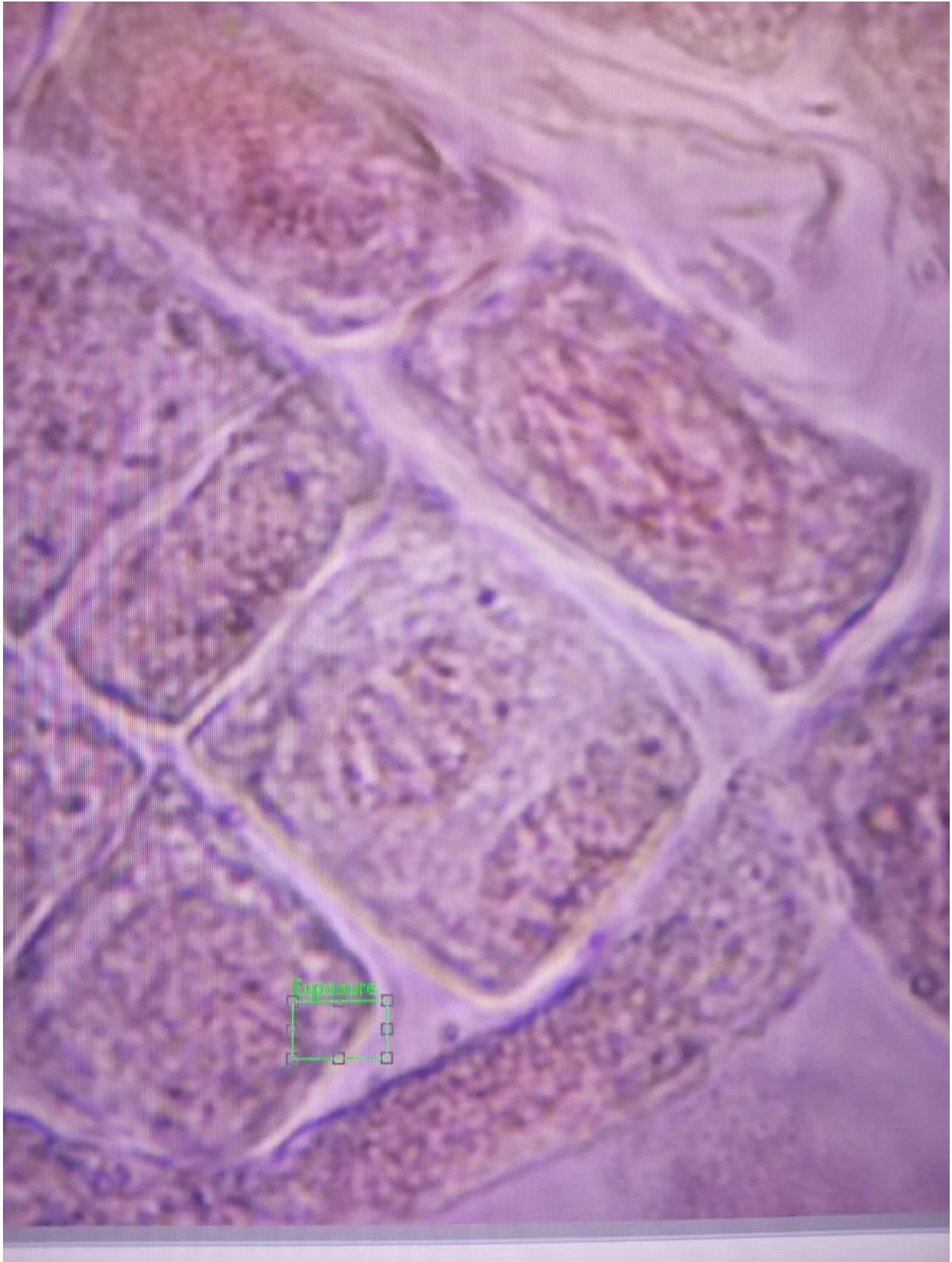
PROPHASE:-



PROPHASE:-



TELOPHASE:-



4) Result (5 marks):

- The meristematic region shows numerous cells in various stages of mitosis
- This confirms that root tips are areas of active growth
- **Interphase:** Longest phase (most cells observed)
- **Prophase:** Most common mitotic stage (longest mitotic phase)
- **Metaphase:** Moderate number of cells
- **Anaphase:** Fewer cells (shortest phase - happens quickly)
- **Telophase:** less number of cells.
- Mitosis occurs in a predictable sequence - All stages can be identified in order
- Cell division is continuous - Multiple stages visible simultaneously
- Growth occurs through cell division - Active mitosis in meristematic tissue
- Phase duration varies - Longer phases show more cells (interphase and prophase)

5) Labster Simulation (5 marks):

DONE

6) Activity (5 Marks):

Click on the link https://www.biologycorner.com/projects/mitosis/onion_root.html and follow the instructions. Please view each highlighted area in sequence (left to right) and indicate the phase of mitosis and write the name of each phase shown below.

- Highlighted area/Picture 1: Anaphase
- Highlighted area/Picture 2: Metaphase
- Highlighted area/Picture 3: Prophase
- Highlighted area/Picture 4: Telophase
- Highlighted area/Picture 5: Cytokinesis

7) Problems:

Staining was not for long enough so that's why cell division was not clear under microscope.

Microscope was not good enough/ Stain was not the best.